

Chapter 5: ML for Casual Inference

5.1 Introduction to Tree-Based Methods

- Decision trees are **non-parametric supervised learning methods** used for classification and regression.
- A decision tree recursively partitions the covariate space into relatively homogeneous subgroups.
- The resulting tree consists of:
 - **Root node** – starting point of the tree.
 - **Internal nodes** – decision/splitting rules.
 - **Branches** – paths defined by splitting rules.
 - **Terminal nodes/leaves** – final subgroups used for prediction.
- Depending on the outcome:
 - **Classification trees** are used for discrete outcomes.
 - **Regression trees** are used for continuous outcomes.
- **CART (Classification and Regression Trees)** is an umbrella term for classification and regression tree procedures.

Important distinction for causal inference

Conventional CART is primarily a **prediction method**:

$$Y=f(X)$$

where X represents covariates and Y represents the outcome.

In causal inference, the objective changes from predicting the outcome to estimating the **effect of treatment**:

$$\tau(X)=E[Y(1)-Y(0)|X]$$

Therefore, causal tree methods modify the tree-building procedure so that the tree identifies **subgroups with different treatment effects**, rather than simply subgroups with different outcome values.

5.2 Why Use Tree-Based Models for Causal Inference?

Traditional regression-based causal models often require assumptions about the functional form of relationships.

For example:

$$Y=\beta_0+\beta_1W+\beta_2X+\epsilon$$

This may not adequately capture:

- nonlinear relationships,
- interactions,

- threshold effects,
- complex relationships between treatment and covariates.

Tree-based methods can learn these patterns directly from the data.

Main advantages

- Non-parametric or highly flexible modelling.
- Can capture **nonlinear relationships**.
- Can detect **interactions** between covariates.
- Can identify **heterogeneity in treatment effects**.
- Can accommodate many predictors.
- Require less specification of the functional form than conventional parametric models.

5.3 Conventional CART versus Causal Trees

This distinction should be one of the most important points in your chapter.

Conventional CART

The objective is to find partitions that improve **outcome prediction**.

For example:

Which characteristics divide patients into groups with different mortality?

Causal Tree

The objective is to find partitions that produce groups with different **treatment effects**.

For example:

Which patient characteristics divide patients into groups in which the treatment has different effects?

So:

Conventional CART	Causal Tree
Predicts outcome	Estimates treatment effect
$Y=f(X)$	$\tau(X)=E[Y(1)-Y(0) X]$
Splits based on outcome prediction	Splits based on treatment-effect heterogeneity
Primarily predictive	Causal inference

Does not automatically estimate causal effects Specifically designed for treatment-effect estimation

5.4 Causal Trees

A causal tree is a tree-based approach designed to identify **heterogeneous treatment effects**.

Suppose:

- W = treatment
- Y = outcome
- X = baseline covariates

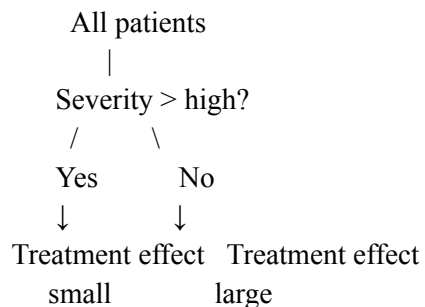
The causal tree attempts to divide individuals into subgroups where:

$$E[Y(1) - Y(0) | X]$$

differs between groups.

Example

Suppose the treatment effect differs according to disease severity:



The tree therefore provides information about **who benefits more or less from treatment**.

This is called: **Treatment-effect heterogeneity** Or **Heterogeneous treatment effects**.

5.5 Conditional Average Treatment Effect (CATE)

A major quantity estimated by causal trees is the **Conditional Average Treatment Effect**:

$$CATE(x) = E[Y(1) - Y(0) | X=x]$$

It answers:

What is the expected treatment effect among individuals with characteristics $X=x$?

This is different from the overall:

$$ATE = E[Y(1) - Y(0)]$$

because the ATE gives one average effect for the entire population, whereas CATE allows the effect to vary across patient characteristics.

5.6 Honest Estimation

One important feature of causal trees is **honest estimation**.

Problem

If the same observations are used to:

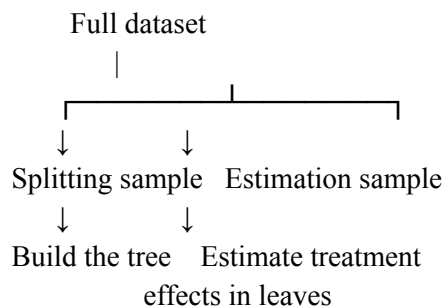
1. Find subgroups with apparently different treatment effects, and
2. Estimate the treatment effect within those subgroups,

the model can overfit.

The treatment effects may appear different simply because of random variation.

Honest approach

The data are divided into different samples:



Thus:

- **Splitting sample** → determines the tree structure.
- **Estimation sample** → estimates treatment effects within the resulting subgroups.

Key point

Honesty separates subgroup discovery from treatment-effect estimation and reduces overfitting.

Your source specifically identifies this as a major difference between conventional CART and causal tree methodology.

5.7 Bayesian Additive Regression Trees (BART)

Concept

BART = Bayesian Additive Regression Trees.

Unlike a conventional CART model that uses one tree, BART uses an **ensemble of many trees**.

Each individual tree is deliberately kept relatively weak, while the combined collection of trees provides a flexible model.

Conceptually:

$$Y = g(X; T_1, M_1) + g(X; T_2, M_2) + \dots + g(X; T_m, M_m) + \epsilon$$

where:

- T_j = structure of tree j
- M_j = terminal-node parameters
- $g(\cdot)$ = prediction generated by the tree
- ϵ = error term.

The trees are combined to model the outcome flexibly.

5.8 Why BART is Useful for Causal Inference

BART can model the relationship:

$$Y = f(W, X) + \epsilon$$

where:

- W = treatment
- X = confounding covariates
- Y = outcome.

Unlike a simple linear regression model, BART does not require the researcher to specify a particular linear relationship between treatment, covariates and outcome.

It can capture:

- nonlinear effects,
- interactions,
- complex relationships,
- high-dimensional covariate structures.

Potential outcomes

Once the model estimates the outcome under each treatment condition, it can be used to estimate:

$$Y^i(1)$$

and

$$Y^i(0)$$

and therefore:

$$ITE_i = Y^i(1) - Y^i(0)$$

The individual treatment effects can then be aggregated to estimate the ATE.

5.9 BART and Treatment-Effect Heterogeneity

BART can be used to estimate:

Average Treatment Effect

$$ATE = E[Y(1) - Y(0)]$$

Individual Treatment Effect

$$ITE_i = Y_i(1) - Y_i(0)$$

Conditional Average Treatment Effect

$$CATE(x) = E[Y(1) - Y(0) | X=x]$$

Therefore, BART can be useful when the research question is not simply:

"Does the treatment work?"

but:

"For whom does the treatment work, and how does the treatment effect vary across individuals?"

5.10 Random Forests and Causal Forests

A **Random Forest** is an ensemble of multiple decision trees.

Instead of relying on one tree, many trees are generated using:

- bootstrap samples of the data,
- randomly selected subsets of predictors.

The individual trees are then combined.

Conventional Random Forest

The primary objective is prediction:

$$Y=f(X)$$

Causal Forest

The framework can be extended to estimate heterogeneous treatment effects:

$$\tau(X)=E[Y(1)-Y(0)|X]$$

Thus, causal forests use the flexibility of forests to identify variation in treatment effects across individuals or subgroups.

5.11 Trees as Adaptive Nearest-Neighbor Methods

An interesting concept from the source is that **trees and forests can be viewed as adaptive nearest-neighbor methods**.

In ordinary nearest-neighbor methods, two individuals are considered similar according to a predefined distance.

Tree-based methods instead define similarity according to the tree structure.

Individuals who fall into the **same terminal node** are considered similar according to the learned partition.

Therefore:

The tree creates an **adaptive neighborhood** based on the covariate patterns learned from the data.

This can be particularly useful when the outcome changes rapidly in some directions of the covariate space but slowly in others.

5.12 Tree-Based Methods and Heterogeneous Treatment Effects

This should be a dedicated subsection because it is one of the main reasons tree-based causal methods are important.

Traditional causal analysis often focuses on:

ATE

But the treatment effect may differ between individuals:

$$\tau_i = Y_i(1) - Y_i(0)$$

Tree-based methods attempt to identify systematic variation in:

$$\tau(X)$$

For example:

Patient characteristic	Estimated treatment effect
Young, low severity	Large
Young, high severity	Moderate
Older, low severity	Small
Older, high severity	Very small

The tree/forest attempts to discover these patterns **from the data rather than requiring all subgroups to be specified in advance.**

5.13 Summary of Tree-Based Causal Methods

Method	Main idea	Main objective
CART	Single decision tree	Outcome prediction
Causal Tree	Tree specifically optimized for treatment-effect heterogeneity	CATE/subgroup effects
Honest Causal Tree	Separate samples for splitting and estimation	More reliable subgroup effect estimation
BART	Bayesian sum of many trees	Flexible outcome and treatment-effect estimation
Random Forest	Ensemble of randomized trees	Flexible prediction
Causal Forest	Forest adapted for treatment-effect estimation	Heterogeneous treatment effects

5.14 Advantages

Tree-based causal methods have several advantages:

- **Flexible functional form**
 - Do not require a predefined linear relationship.
- **Nonlinearity**

- Can capture nonlinear relationships between covariates and outcomes.
- **Interactions**
 - Can automatically identify interactions.
- **Heterogeneous treatment effects**
 - Can identify subgroups with different treatment effects.
- **High-dimensional covariates**
 - Can handle a relatively large number of predictors.
- **Interpretability**
 - Individual trees can provide intuitive subgroup rules.
- **Reduced model specification**
 - Less need to manually specify interactions and nonlinear terms.

5.15 Limitations

You should also include the limitations.

- Individual trees can be **unstable**.
- Small changes in the data may produce different tree structures.
- Very flexible models can **overfit**.
- Interpretation becomes more difficult for ensembles such as BART and causal forests.
- Treatment-effect estimation still depends on the underlying causal assumptions.
- Tree-based methods do **not eliminate confounding automatically**.
- Results can be sensitive to sample size and the available information about treated and untreated individuals.
- Identifying treatment-effect heterogeneity can be particularly difficult when the sample is small.

Very important

Using a machine-learning model does not by itself make an analysis causal.

The causal interpretation still depends on assumptions such as:

- consistency,
- exchangeability/no unmeasured confounding,
- positivity.

So the tree is the **estimation tool**; it does not replace the causal identification framework.